

Ernest Choi

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Projects

AI Resume & Job-Search Automation Bot | *Python*

github.com/ErChoi-com/DiscordBot

- Designed an **LLM**-driven resume and cover-letter pipeline with fallback across Google Gemini, Groq, and OpenRouter, prompting for structured **JSON** output backed by **SQLite** storage and a pytest test suite.
- Built a **Discord.py** bot in **Python** that scrapes and deduplicates job postings through **REST APIs** and **Playwright** browser automation, scheduling the work with **asyncio** so interactive commands preempt background scraping.

Bookstore App | *Java, JavaFX*

- Created a bookstore app in **Java** around OOP design principles, with persistent file-backed record storage.
- Built a responsive **JavaFX** interface and wired it to the app's backend functions.

Web Messaging App | *JavaScript, Node.js, Python, SQL*

ernestchoi.com/terminalcards

- Built a collaborative web application in **JavaScript** and **Node.js**, utilizing **WebRTC** for real-time messaging and file sharing.
- Implemented **Python**-driven autocomplete and in-browser code execution for sophisticated user applications.

Experience

Frontend Development Intern, MCG3D – Remote

January 2026 – March 2026

- Built production web applications in **React** and **TypeScript**, keeping the component architecture scalable and the interface fast and user-centric.
- Integrated frontend interfaces with **Python Flask** services, utilizing AI tools to accelerate development and feature enhancement.
- Deployed and tested containerized **Docker** applications to ensure stability across engineering systems.

Machine Learning Intern, Goppter Commerce Solutions – Remote

September 2025 – December 2025

- Engineered a distributed data pipeline for **Python** services using **FastAPI** and **Kafka** to process enterprise financial data.
- Built predictive models in **TensorFlow** to generate adaptive signals for quantitative trading applications.
- Deployed analytics dashboards in **Tableau** and **Power BI**, surfacing real-time predictive insights from models trained with **PyTorch Lightning** and NumPy.

Technical Development Intern, City of Markham – Markham, Ontario

July 2022 – August 2024

- Validated and tested system changes against quality standards and functional requirements, reducing reported errors by 20%.
- Maintained clear **LaTeX** documentation of website workflows and operating processes, helping cut website downtime by 41%.
- Tracked project milestones and reported progress to management in **Microsoft Suite**, reducing coordination delays by 14%.

Skills

Languages: Java, Python, JavaScript/Node.js, C++, C, SQL, HTML, CSS

Frameworks/Libraries: OpenGL, WebRTC, JavaFX, React, TypeScript, Flask, FastAPI, TensorFlow

Tools: Git, PostgreSQL, Ubuntu Linux, Docker, Kafka, Tableau, Power BI, FPGA

Platforms: GitHub, GitLab, Raspberry Pi, Arduino

Relevant Courses: Digital Systems, Microprocessor Systems, Object Oriented Eng Analysis and Design, Electronic Circuits I

Education

Toronto Metropolitan University – Bachelor of Engineering in Computer Engineering

2023–2027